THE CATEGORY OF COMPACTIFICATIONS OF A COMPACTIFYABLE SCHEME IS COFILTERED, WITH THE SCHEME THEORETICALLY DENSE COMPACTIFICATIONS INITIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. One step of part (b) is flagged by the author as omitted and is kept verbatim in the delivered proof (source lines 9369-9371): "follows from the first (we omit the solution to this categorical exercise)". Part (a) and part (c) are written out in full inside the counted span; the omitted step is the formal consequence of the just-proved cofilteredness recorded in the statement.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ATU (sources/stacks-flat/pinned-original-source.tex line 9316).

Allowed standard prerequisites (editorial).

- Cofiltered (codirected) categories and initial objects and subcategories. Categories, Definitions labelled categories-definition-codirected (Stacks tag 04AZ) and categories-definition-initial (Stacks tag 09WP), invoked at statement line 9325 and proof lines 9334, 9366. Supply the three conditions that cofilteredness amounts to and the two conditions for a full subcategory to be initial; checking those conditions for the category of compactifications is the content of the counted span.
- Scheme theoretic closure and scheme theoretic image of a partial section. Morphisms, Lemma labelled morphisms-lemma-quasi-compact-immersion (Stacks tag 01RG) and Lemma labelled morphisms-lemma-scheme-theoretic-image-of-partial-section (Stacks tag 0CNG), invoked at proof lines 9364 and 9376. Supply that X is dense and scheme theoretically dense in its scheme theoretic closure, and the topological description of the preimage of j(X) under f; the construction of the dominating compactification and of the equalizer is inside the counted span.

Source reference. Original definition 09WP: sources/stacks-chow/tags.txt, source lines 12502–12502. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9366 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original definition 04AZ: sources/stacks-chow/tags.txt, source lines 5634–5634. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9334 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01RG: sources/stacks-chow/tags.txt, source lines 2301–2301. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9364 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CNG: sources/stacks-chow/tags.txt, source lines 15891–15891. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9377 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let X be a compactifyable scheme over S.*

(a) *The category of compactifications of X over S is cofiltered.*
(b) *The full subcategory consisting of compactifications $j : X \to \overline{X}$ such that $j(X)$ is dense and scheme theoretically dense in $\overline{X}$ is initial (Categories, Definition 09WP).*
(c) *If $f : \overline{X}' \to \overline{X}$ is a morphism of compactifications of X such that $j'(X)$ is dense in $\overline{X}'$, then $f^{-1}(j(X)) = j'(X)$.*

Proof. To prove part (a) we have to check conditions (1), (2), (3) of Categories, Definition 04AZ. Condition (1) holds exactly because we assumed that X is compactifyable. Let $j_i : X \to \overline{X}_i$, $i = 1, 2$ be two compactifications. Then we can consider the scheme theoretic image $\overline{X}$ of $(j_1, j_2) : X \to \overline{X}_1 \times_S \overline{X}_2$. This determines a third compactification $j : X \to \overline{X}$ which dominates both j_i:

$$(X, \overline{X}_1) \longleftarrow (X, \overline{X}) \longrightarrow (X, \overline{X}_2)$$

Thus (2) holds. Let $f_1, f_2 : \overline{X}_1 \to \overline{X}_2$ be two morphisms between compactifications $j_i : X \to \overline{X}_i$, $i = 1, 2$. Let $\overline{X} \subset \overline{X}_1$ be the equalizer of f_1 and f_2. As $\overline{X}_2 \to S$ is separated, we see that $\overline{X}$ is a closed subscheme of $\overline{X}_1$ and hence proper over S. Moreover, we obtain an open immersion $X \to \overline{X}$ because $f_1|_X = f_2|_X = \mathrm{id}_X$. The morphism $(X \to \overline{X}) \to (j_1 : X \to \overline{X}_1)$ given by the closed immersion $\overline{X} \to \overline{X}_1$ equalizes f_1 and f_2 which proves condition (3).

Proof of (b). Let $j : X \to \overline{X}$ be a compactification. If $\overline{X}'$ denotes the scheme theoretic closure of X in $\overline{X}$, then X is dense and scheme theoretically dense in $\overline{X}'$ by Morphisms, Lemma 01RG. This proves the first condition of Categories, Definition 09WP. Since we have already shown the category of compactifications of X is cofiltered, the second condition of Categories, Definition 09WP follows from the first (we omit the solution to this categorical exercise).

Proof of (c). After replacing $\overline{X}'$ with the scheme theoretic closure of $j'(X)$ (which doesn't change the underlying topological space) this follows from Morphisms, Lemma 0CNG. □